MARCELA S. MELARA

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RESEARCH INTERESTS

Trustworthy distributed systems, OS security, networks, applied cryptography, hardware-based security

EDUCATION

Princeton University, Princeton, NJ

Ph.D., Computer Science — September 2019
Dissertation: *Intra-Process Least Privilege and Isolation for Emerging Applications*
Advisor: Michael J. Freedman

M.S.E., Computer Science — June 2014
Thesis: *CONIKS: Preserving Secure Communication with Untrusted Identity Providers*
Advisor: Edward W. Felten

Hobart and William Smith Colleges (HWS), Geneva, NY

B.S., summa cum laude, Computer Science — May 2012
Honors Thesis: *ELARA: Environmental Liaison and Automated Recycling Assistant*
Second Major in French and Francophone Studies. Minor in Physics.

PROFESSIONAL EXPERIENCE

Research Scientist, Security & Privacy Research — Oct 2019–present
Intel Labs

Lead research on trustworthy, high-integrity and confidential software supply chain and cloud systems:

- Design systems for attested software build environments using trusted execution.
- Author Supply Chain Attribute Integrity (SCAI) specification (now under CNCF in-toto) for fine-grained software artifact attestation.
- Develop framework for end-to-end machine learning/AI lifecycle integrity and provenance.

Graduate Research Intern — June 2018–Feb 2019
Intel Labs

- Designed a fine-grained memory compartmentalization scheme for Intel SGX using memory protection keys.

OS Security Engineering Intern, CoreOS Team — Summer 2014 & 2015
Apple Inc.

- Evaluated deployment of key transparency as part of a large-scale system.
- Improved sandboxing technology in core operating system.

Technical Intern — November 2010–April 2011
Seneca7 Race Committee

- Designed and implemented a web-based program for real-time tracking of relay race runners.

PUBLICATIONS AND PRESENTATIONS

Refereed Papers & Articles

"Proving Attributes about Confidential Compute Services with Validation and Endorsement Services" A. Vahldiek-Oberwagner, **M. S. Melara**. *IEEE 8th Workshop on System Software for Trusted Execution (SysTEX)*. July 2025.

"Atlas: A Framework for ML Lifecycle Provenance & Transparency" M. Spoczynski, **M. S. Melara**, S. Szyller. *IEEE 8th Workshop on System Software for Trusted Execution (SysTEX)*. July 2025.

"A Viewpoint on Software Supply Chain Security: Are We Getting Lost in Translation?" **M. S. Melara** and S. Torres-Arias. *IEEE Security & Privacy*, Vol. 21, Issue 6. Nov 2023.

"Hardware-Enforced Integrity and Provenance for Distributed Code Deployments" **M. S. Melara** and M. Bowman. *NIST Workshop on Enhancing Software Supply Chain Security*. June 2021.

"CONIKS: Bringing Key Transparency to End Users" **M. S. Melara**, A. Blankstein, J. Bonneau, E. Felten, M. Freedman. *USENIX Security Symposium*. August 2015. Caspar Bowden PET Award, 2017.

"Shining the Floodlights on Mobile Web Tracking — A Privacy Survey" C. Eubank, **M. Melara**, D. Perez Botero, A. Narayanan. *W2SP*, 2013.

"Vireos: an Integrated, Bottom-Up Educational Operating Systems Project with FPGA support" M. Corliss, **M. Melara**. *ACM SIGCSE*, 2011.

Invited Talks

"Securing the AI Lifecycle: Trust, Transparency & Tooling in Open Source" S. Evans, M. Maruseac and **M. Melara**. *OpenSSF Tech Talk*. Sep 2025.

"Understanding the Software Supply Chain Trust Landscape" **M. Melara**. *C2PA Conformance Working Group*. Jul 2024.

"Securing the Software Supply Chain: An In-Depth Exploration of SLSA" M. Lieberman, **M. Melara**, J. Lock, L. Capadan. *OpenSSF Tech Talk*. Oct 2023.

"Building Trust with Attestation" **M. Melara**, V. Scarlata. *Open at Intel Podcast*. May 2023.

"Software Supply Chains" **M. Melara**, B. Domingues. *Open at Intel Podcast*. Mar 2023.

"EnclaveDom: Privilege Separation for Large-TCB Applications in Trusted Execution Environments" **M. Melara**. *Microsoft Research Cryptography & Privacy Colloqium*. Sep 2020.

Conference Talks

"The Whole Is Greater Than the Sum of Its Parts: A Case for Interoperable Supply Chain Tooling" Hayden Blauzvern and **M. Melara** *Open Source SecurityCon*. Nov 2025.

"Harnessing in-toto Attestations for Security and Compliance With Next-gen Policies" **M. Melara** and T. Karthik Kuppusamy *OpenSSF Community Day NA*. Jun 2025.

"Building Trust in ML: Mapping the Model Lifecycle for ML Integrity and Transparency" **M. S. Melara** *Open Source Summit NA*. Jun 2025.

"Auditing the CI/CD Platform: Reproducible Builds vs. Hardware-Attested Build Environments, Which is Right for You?" **M. S. Melara**, C. Kimes. *ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED)*. Oct 2024.

"TPMs, Merkle Trees and TEEs: Enhancing SLSA with Hardware-Assisted Build Environment Verification" **M. Melara**, C. Kimes. *Open Source Summit NA*. Apr 2024.

"Panel Discussion: Improving Supply Chain Integrity with OpenSSF Technologies" A. Le Hors, M. Lieberman, J. White, **M. Melara**, I. Hepworth. *Open Source Summit NA*. Apr 2024.

"Panel Discussion: DEI for the OpenSSF Community" M. McElaney, J. Kjell, J. White, C. Voong, **M. Melara**. *SOSS Community Day NA*. Apr 2024.

"All things in-toto! Supply chain attestations, policies, and adoption stories, oh my!" **M. Melara**, S. Torres-Arias. *KubeCon & CloudNativeCon NA*. Nov 2023.

"Using FPGAs to Create a Complete Computer System for the Classroom" **M. Melara**. *NYCWiC 2011*.

Patents

"Concept for Performing Operations on an Asset." C. M. Bowman, P. Narayana Moorthy, B. Vavala, **M. S. Melara**. *US Patent Application 18/343,797*. 2024.

"Method and apparatus for multi-dimensional attestation for a software application." **M. S. Melara**, B. Vavala, M. Steiner, V. Scarlata, A. L. Vahldiek-Oberwagner. *US Patent Application 18/311,253*. 2023.

"Attestation of operations by tool chains." V. Scarlata, A. Trivedi, R. Lal, **M. S. Melara**, M. Steiner, A. L. Vahldiek-Oberwagner. *US Patent 11650800*. 2023.

"Optimizing deployment and security of microservices." P. Saxena, A. L. Vahldiek-Oberwagner, M. Vij, K. A Doshi, C. H. Morales, C. M. Bowman, **M. S. Melara**, M. Steiner. *US Patent Application 17/561,676*. 2022.

Blog Posts

"New Atlas CLI Open Source Tool Manages Machine Learning Model Provenance and Transparency" M. Spoczynski, **M. Melara**, S. Szyller. *Intel Community Tech Innovation Blog*. Jun 2025.

"Building Trust in AI: An End-to-End Approach for the Machine Learning Model Lifecycle" M. Spoczynski, **M. Melara**, S. Szyller. *Intel Community Tech Innovation Blog*. Dec 2024.

"The Opportunity for DEI Participation in the Security Industry (And OpenSSF)" C. Voong, J. White, J. Kjell, **M. Melara**, M. McElaney. *OpenSSF Blog*. May 2024.

"Why Making Johnny's Key Management Transparent is So Challenging" **M. Melara**. *Freedom to Tinker*. March 2016.

"There's Something Wrong With This Picture..." **M. Melara**. Guest blogger, *Grand Central Blog*. November 2010.

"Busy Moms Need Energy" **M. Melara**. Guest blogger, *Grand Central Blog*. October 2010.

Posters

"Protecting the IoT Against Data Leaks through Intra-Process Access Control" **M. Melara**. *Stony Brook University, National Security Institute Security & Privacy Day 2017*.

"Building an Automatic and Scalable Tool for Improving Environmental Recycling: ELARA" **M. Melara**. *HWS Summer Research Symposium 2011*.

"Using FPGAs to Create a Complete Computer System for the Classroom" **M. Melara**. *HWS Summer Research Symposium 2010*.

Public Manuscripts

"Trustworthy and Confidential SBOM Exchange" E. Abu Ishgair, Chinenye Okafor, **M. S. Melara** and S. Torres-Arias. *ArXiv Preprint*. Sep 2025.

"Enabling Security-Oriented Orchestration of Microservices" **M. S. Melara** and M. Bowman. *ArXiv Preprint*. May 2021.

"EnclaveDom: Privilege Separation for Large-TCB Applications in Trusted Execution Environments" **M. S. Melara**, M. J. Freedman, and M. Bowman. *ArXiv Preprint*. July 2019.

"Pyronia: Redesigning Least Privilege and Isolation for the Age of IoT" **M. S. Melara**, D. Liu, and M. J. Freedman. *ArXiv Preprint*. March 2019.

MENTORING

PhD Advising

Eman Abu Ishgair (*Purdue University*). With Prof. Santiago Torres Arias

Yaxuan "Alice" Wen (*New York University*). With Prof. Justin Cappos

Interns Mentored, Intel Labs

Eman Abu Ishgair (*Purdue University*), Summer 2024
Project: "Trustworthy and Confidential SBOM Exchange"

Undergraduate Students Mentored

Jessica May. CRA-W Collaborative Research Experiences for Undergraduates, Summer 2017
Project: "Stormship: Smart Tool for Revealing Malicious Scripts Hidden in Plain Sites". Co-advised with Nick Feamster

Huy Quoc Vu. Google Summer of Code, Summer 2016
Project: "CONIKS for Tor Messenger". Co-mentored with Arlo Breault (*The Tor Project*)

Michael Rochlin. Princeton University Junior Independent Work, Spring 2015
Project: "Coniks 2.0: Publicly Verifiable Keystore with Key Changing and Verifying Capabilities". Co-advised with Ed Felten

Intel-Princeton REU

Lois Omotara, Summer 2023
Priya Naphade, Summer 2022

Científico Latino Graduate School Mentoring Initiative

Jessica Williams, Fall 2023
Brittany Gomez, Fall 2020

TEACHING

Graduate Teaching Fellow Summer 2017–Winter 2018
Princeton University, McGraw Center for Teaching and Learning
- Provided instructional consultation for TAs via classroom observations.
- Led the annual Assistant in Instruction Orientation for new Computer Science TAs.

Assistant in Instruction
Princeton University
- COS 461: Computer Networks Spring 2014
  Instructor: Prof. Michael Freedman
- COS 318: Operating Systems Fall 2013
  Instructors: Prof. Kai Li, Dr. Andrew Bavier

Physics Teaching Fellow Fall 2011–Spring 2012
Hobart and William Smith Colleges, Center for Teaching and Learning
- Held walk-in evening office hours for students taking Physics courses of any level.

Evening Teaching Assistant Fall 2010–Spring 2011
Hobart and William Smith Colleges, Dept. of Mathematics and Computer Science
- Held walk-in evening office hours for students taking introductory Computer Science courses.

Teaching Assistant
Hobart and William Smith Colleges, Dept. of Mathematics and Computer Science
- CPSC 124: Introduction to Programming Fall 2011, Spring 2012
  Instructors: Prof. Davd Eck, Prof. Carol Critchlow, respectively
- MATH 131: Calculus II Fall 2009
  Instructor: Prof. Kevin Mitchell

HONORS AND AWARDS

Open Source Security Foundation Golden Egg Award June 2025

USENIX Security Noteworthy Reviewer Award August 2023

Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies July 2017

Siebel Scholars Class of 2014 July 2013

Princeton University President's Fellowship Fall 2012–Spring 2013

Phi Beta Kappa May 2012

Honors in Computer Science April 2012

HWS Dept. of Mathematics and Computer Science John S. Klein Prize April 2012

Roderic '52 and Patricia '53 Ross Endowed Centennial Scholarship Fall 2011–Spring 2012

Hai Timiai Women's Senior Honors Society April 2011

HWS Dept. of Mathematics and Computer Science William Ross Proctor Prize April 2010

Phi Beta Kappa Book Award April 2010

First Year Academic Achievement Award April 2009

William Smith College Dean's List Fall 2008–Spring 2012

SERVICE ACTIVITIES

Co-Organizer, Linux Foundation Open Robust Compartmentalization Alliance (ORCA) 2025–present

Technical Advisory Council, Open Source Security Foundation (OpenSSF) 2024–present

Co-Chair, OpenSSF Belonging, Empowerment, Allyship and Representation Working Group 2024–present

Workshop Organizer, ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses (SCORED) 2022–present

Specification Maintainer, CNCF in-toto Attestation Framework 2023–present

Specification Maintainer, OpenSSF Supply-chain Levels for Software Artifacts (SLSA) 2024–present

Program Committee, Open Source SecurityCon NA 2025

Program Committee, OpenSSF Community Day NA 2025

Program Committee, ACM EuroSec 2024–2025

Program Committee, Linux Foundation SigstoreCon: Supply Chain Day 2024

Program Committee, Linux Foundation SOSS Community Day EU 2024

External Reviewer, Proceedings on Privacy Enhancing Technologies (PoPETS) 2018

External Reviewer, International World Wide Web Conference (WWW) 2015

Regional Lead, NJ, Siebel Scholars Fall 2014–Spring 2016

Faculty Hiring Committee, HWS Dept. of Physics Spring 2012

Faculty Hiring Committee, HWS Dept. of French and Francophone Studies Spring 2012